NOT MEASUREMENT SENSITIVE
MIL-P-85856/16
7 December 1988

# MILITARY SPECIFICATION SHEET

## POWDER MATERIALS FOR FLAME AND PLASMA SPRAYED COATINGS MOLYBDENUM, TYPE I, CLASS 1, COMPOSITION P

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the powder described herein shall consist of this document and the latest issue of MIL-P-85856.

### Chemical Composition

| Mo | O | Others (1) | |
|---|---|---|---|
| | | Each | Total |
| 99.5 min. | 0.2 max. | - | 0.5 max. |

(1) Powder particles shall be essentially oxide-free.

### Particle Size/Distribution

| | |
|---|---|
| -325 Mesh (-45 µm) | 100.0 min. |
| -20 µm | 80.0 min. |
| -5 µm | 50.0 max. |

### Sprayed Coating Properties

| | |
|---|---|
| Bond Strength | 41.0 MPa (6000 psi) min. |
| Hardness | Rockwell 15N, 75 min. |
| Microstructure | See paragraph 4.5.4.1.2<br>Porosity - 3% max.<br>Unmelted Particles - 10% max. |

FSC 3439

AMSC N/A
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Deposit Efficiency and Deposition Rate

65% min for $N_2/H_2$ @ 8.0 lb/hr
75% min for $A/H_2$ @ 8.0 lb/hr

Comparable Documents

Powder which meets the requirements specified is qualified under the following specifications.

| | |
|---|---|
| Pratt & Whitney Aircraft | PWA 1338 |

Custodians:
Air Force - 20
Navy - AS
Army - MR

Review activities:
Air Force - 99, 71
Navy - SH

Preparing activity:
Navy - AS
(Project No. 3439-0663/16)



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